

Interplay of dimerization and quasiperiodicity in the superconducting proximity effect of a one-dimensional hybrid ring

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Recent studies of the superconducting proximity effect in quasicrystalline and topological systems have opened up a new research direction for exploring how quasiperiodicity and topology influence proximity induced superconductivity. In this work, we investigate spatial variation of the proximity induced pairing amplitude in a hybrid ring composed of a spin singlet superconductor and a normal region described by three different lattice models using the self-consistent Bogoliubov-de Gennes formalism. We first consider the normal region described by the diagonal Aubry-André-Harper (AAH) model. Increasing the quasiperiodic potential enhances spatial fluctuations in the induced order parameter, while progressively suppressing its magnitude in the normal region. Beyond the localization transition, proximity-induced pairing is strongly diminished due to the localized nature of the underlying electronic states. The normal region is then modeled by the Su-Schrieffer-Heeger (SSH) chain to investigate the effect of hopping dimerization. Weak dimerization introduces oscillatory modulations in the induced pairing that extend deep into the normal region, whereas strong dimerization confines these oscillations and significantly reduces the penetration of superconducting correlations. Finally, we study the combined SSH-AAH model to explore the interplay between the dimerized hopping and quasiperiodicity. The results show that the SSH dimerization determines the oscillatory behavior and penetration of the induced pairing, while the AAH potential enhances spatial inhomogeneity and further reduces its magnitude. Together, these two effects provide a versatile means of controlling proximity-induced superconductivity in quasiperiodic hybrid systems.

I. INTRODUCTION

The superconducting proximity effect in hybrid structures comprising superconducting and normal-conducting regions has long been recognized as a remarkable consequence of quantum coherence in mesoscopic systems. When a superconductor¹ is placed in contact with a non-superconducting material, Cooper pairs can penetrate through the interface^{2–5} and induce pairing correlations in the adjacent material. The extent of this induced superconductivity depends strongly on whether its electronic states are extended, critical, or localized. If the underlying wave functions are extended, superconducting correlations can propagate deep into the normal region. In contrast, localized states tend to inhibit this penetration. As a result, the proximity effect provides a sensitive indicator of the electronic properties of the non-superconducting systems^{6–9} and offers a useful route for exploring unconventional electronic phases.

The nature of electronic states is closely related to the ordering of the underlying lattices. In periodic crystals, translational symmetry leads to Bloch states that extend throughout the system and support metallic transport. However, in random disordered one-dimensional (1D) systems, quantum interference causes Anderson localization¹⁰, resulting in insulating behavior. In this context, the quasicrystals lie between these two paradigms^{11,12}. Although they possess a long-range order, they lack translational symmetry. Consequently, their electronic states are generally neither fully extended nor exponentially localized. Instead, they often exhibit critical behavior characterized by self-similar and multifractal patterns^{13–17}. This unique nature of quasiperiodic systems accounts for a diverse range of physical phenomena that are absent in both perfectly ordered crystals and ran-

domly disordered materials.

A paradigmatic model capturing these features is the Aubry-André-Harper (AAH) lattice model^{18,19}, in which a quasiperiodic modulation of the on-site potential with irrational period drives a sharp transition at a finite critical strength in the thermodynamic limit. Below this threshold all eigenstates are extended, above it they are all exponentially localized, and precisely at the critical modulation strength they are self-similar and multifractal. This transition arises from an exact self-duality of the model and is therefore fundamentally distinct from Anderson localization, where arbitrarily weak disorder localizes states in one-dimension. The tunability of the AAH model for different electronic states has motivated extensive generalizations, including models with mobility edges and extended critical phases^{20–25}. The different nature of electronic states of the AAH model motivates researchers to investigate how the induced pair correlation is affected by these states in a superconducting-quasiperiodic system.

Another well-known model is the Su-Schrieffer-Heeger (SSH) chain^{26–29}, a one-dimensional dimerized lattice where the hopping amplitudes take alternating values of t_A and t_B . The SSH model is the prototypical example of a one-dimensional symmetry-protected topological (SPT) insulator³⁰. Its topological phase, realized when $t_A < t_B$, hosts zero-energy edge modes at both ends of an open chain³¹. The interplay between the topological structure of the SSH chain and superconducting correlations has attracted considerable interest because the combination of nontrivial band topology and pairing can give rise to topological superconducting phases^{32–38}.

In recent years, the interplay between quasiperiodicity and superconductivity has attracted considerable attention from both theorists and experimentalists^{39–48}. For 1D Fibonacci quasicrystals, in which hopping amplitudes

follow a Fibonacci substitution sequence with two distinct values, investigations have addressed the proximity effect^{8,9}, Josephson transport^{39,40}, and topological superconducting phases^{42,43,47,48}. Recent studies have shown an enhancement of intrinsic superconductivity in quasiperiodic systems^{42,43,49}, pointing to a deep connection between quasiperiodicity and superconducting pairing analogous to the multifractal enhancement of the critical temperature in disordered systems. Beyond 1D systems, superconductivity has been studied extensively in two-dimensional quasiperiodic lattices based on Penrose and Ammann-Beenker tilings^{45,46}, where the superconducting gap and local pairing amplitude exhibit a self-similar and highly non-uniform spatial pattern governed by the underlying quasiperiodic geometry. An important experimental work was the observation of superconductivity in an Al-Zn-Mg quasicrystal⁴⁸, and more recently superconducting behavior was also noticed in a van der Waals layered dodecagonal quasicrystal⁵⁰, establishing that quasiperiodic order can support genuine superconducting phases.

The superconducting proximity effect in quasiperiodic and disordered systems studied by Rai^{8,9} showed that spatial fluctuations in the proximity-induced order parameter increase with decreasing hopping ratio, i.e., with increasing quasiperiodic modulations and their results establish a direct connection between the localization properties of the single-particle states and the spatial structure of the proximity-induced pairing, motivating a systematic study of this correspondence in other quasiperiodic models. The present work is devoted to a systematic investigation of the superconducting proximity effect in a 1D hybrid ring, in which the normal region is successively replaced by two qualitatively different models of quasiperiodic and topological order. We self-consistently compute the spatial distribution of the order parameter using the Bogoliubov-de Gennes (BdG)^{55,56} framework at zero temperature.

In the first part of this work, the normal region is described by the AAH model. We explore how the proximity-induced pair correlations at a given lattice site in the normal region evolve as the quasiperiodic strength is tuned across the localization transition. In the extended phase, Cooper pair correlations penetrate deeply into the normal region. As localization sets in, the penetration is progressively suppressed. This provides a direct, real-space signature of the AAH localization transition in a proximity-effect measurement. To characterize the sensitivity of the proximity effect to the specific configuration of the onsite AAH modulation, we analyze the variation of the self-consistently determined order parameter Δ_i at a fixed site i in the interior of the normal region for different values of the AAH phase ϕ . Keeping the chain length L_N of the normal region fixed, varying $\phi \in [0, 2\pi]$ generates a continuous family of AAH configurations, each characterized by a distinct on-site quasiperiodic potential landscape with fixed quasiperiodic modulation strength. Although all such configurations generated by varying the phase factor ϕ share the same statistical properties in the thermodynamic limit, they differ locally in their on-site energy landscapes for

any finite L_N , leading to sample-to-sample fluctuations in the proximity-induced pairing amplitude. For every choice of the phase ϕ , the BdG equations are iteratively solved until a self-consistent solution is obtained. We then extract the superconducting order parameter at representative sites chosen within the normal segment. By repeating this procedure for different values of ϕ , we obtain the phase dependence of Δ_i . This may provide a

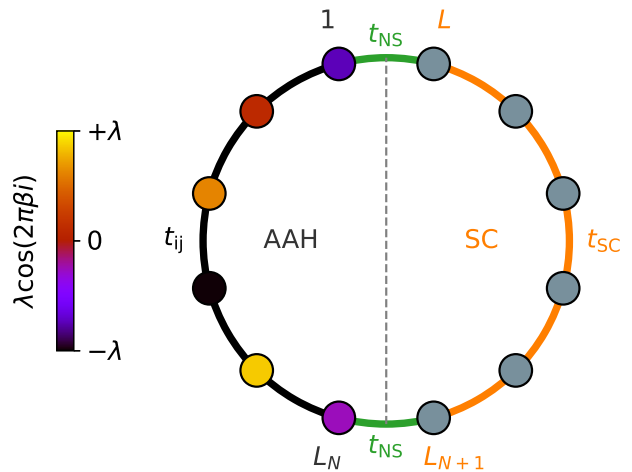


FIG. 1: (Color online.) Schematic representation of the 1D hybrid ring, consisting of a normal (N) segment of length L_N and a superconducting (SC) segment of length $L_{SC} = L - L_N$. The normal region is characterized by nearest-neighbor hopping amplitude t_{ij} together with a quasiperiodic onsite potential of strength λ_{AAH} , while the superconducting region is described by a uniform hopping amplitude t_{SC} . In the normal region, different colored sites correspond to sites with correlated potentials, as indicated by the color bar. The N and SC regions are coupled through normal-superconductor interfaces, represented by the dashed vertical line, with a hopping amplitude t_{NS} .

direct measure of how sensitive the proximity-induced superconductivity is to changes in the quasiperiodic potential landscape while keeping the quasiperiodic modulation strength λ_{AAH} constant.

We next turn to the normal region described by the SSH-AAH model. In this system, the alternating hopping amplitudes of the SSH chain coexist with the quasiperiodic on-site modulation of the AAH model. The SSH-AAH normal region therefore provides a convenient platform for studying how the superconducting proximity effect is influenced by both topological band structure and quasiperiodicity. Our analysis is carried out in two stages. We begin with $\lambda_{AAH} = 0$, where the normal region is reduced to the SSH chain. In this limit, we investigate how the proximity-induced pairing evolves as the system is tuned across the SSH topological phase transition by varying the ratio t_A/t_B . We then introduce a finite quasiperiodic potential and inspect how it modifies the superconducting correlations established in the topological phase. Particular attention is paid to the competition between the robustness associated with hopping

dimerization and the localization tendencies induced by quasiperiodicity.

The rest part of the work is follows. In Sec. II we describe the Hamiltonian of the hybrid ring, the BdG mean-field decoupling method, and the self-consistent numerical procedure for calculating the results. Section III presents all the numerical results, organized into three sub-sections, covering the AAH region (III A), the SSH chain (III B), and the combined SSH-AAH model (III C). Section IV includes the essential findings of our work, an outlook for future work, and possible experimental realizations of our proposed model hybrid quantum ring.

II. MODEL HYBRID QUANTUM RING AND THE METHOD

A. Hamiltonian of the Hybrid Ring

We consider a hybrid ring of L total sites (see Fig. 1), consisting of $L_N = L - L_{SC}$ normal sites hosting AAH type quasiperiodic potential and L_{SC} superconducting sites. Throughout this work, the lattice spacing is set to unity, so that the lengths are measured in lattice units. The full Hamiltonian of the hybrid ring is written as

$$H = H_N + H_{SC} + H_{NS}, \quad (1)$$

where H_N , H_{SC} , and H_{NS} describe the Hamiltonians of the normal, superconducting, and the normal-superconductor (NS) interface region, respectively.

Normal region: The normal region is described by the SSH-AAH tight-binding (TB) Hamiltonian

$$H_N = - \sum_{\langle i,j \rangle \in N} t_{ij} c_i^\dagger c_j + \lambda_{AAH} \sum_{i \in N} \cos(2\pi\beta i + \phi) c_i^\dagger c_i, \quad (2)$$

where $c_i^\dagger$ (c_i) creates (annihilates) an electron at site i , λ_{AAH} is the quasiperiodic modulation strength, $\beta = (\sqrt{5} - 1)/2$ is the inverse golden ratio, and ϕ is the AAH phase. The hopping amplitude t_{ij} takes the form

$$t_{ij} = \begin{cases} t_A & \text{if } (i, j) \text{ is an intra-cell bond,} \\ t_B & \text{if } (i, j) \text{ is an inter-cell bond.} \end{cases} \quad (3)$$

The Hamiltonian given in Eq. (2) represents three different models depending on the choice of the TB parameters. Here, we represent them one by one as follows.

- *AAH model:* Setting $t_A = t_B \equiv t_N$ makes the hopping uniform throughout the normal region, and Eq. (2) corresponds to the well known AAH model with a quasiperiodic on-site potential of strength λ_{AAH} .
- *SSH model:* Setting $\lambda_{AAH} = 0$ removes the on-site potential entirely, and Eq. (2) reduces to the SSH chain with alternating bonds t_A and t_B . The SSH chain is topologically non-trivial for $t_A < t_B$, whereas it is topologically trivial for $t_A > t_B$ with the topological phase transition occurring at $t_A = t_B$ where the bulk gap $E_{\text{gap}} = 2|t_B - t_A|$ closes.

- *SSH-AAH model:* For the general case where $t_A \neq t_B$ and $\lambda_{AAH} \neq 0$, Eq. (2) corresponds to the SSH-AAH chain, in which the dimerized hopping structure of the SSH chain coexists with quasiperiodic on-site disorder. This model allows us to study the competition between hopping dimerization and quasiperiodic localization and their combined effect on the superconducting proximity effect.

The above-mentioned three limiting cases are summarized in Table I.

TABLE I: Three models described by the generalized normal Hamiltonian Eq. (2).

Model	t_A/t_B	λ_{AAH}
AAH	$t_A = t_B \equiv t_N$	$\neq 0$
SSH	$\neq 1$	$= 0$
SSH-AAH	$\neq 1$	$\neq 0$

Superconducting region: The superconducting region is described by the following Hamiltonian¹

$$H_{SC} = -t_{SC} \sum_{\substack{i,j \in SC \\ \langle i,j \rangle, \sigma}} c_{i\sigma}^\dagger c_{j\sigma} + V_{SC} \sum_{i \in SC} c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger c_{i\downarrow} c_{i\uparrow}, \quad (4)$$

where t_{SC} is the nearest-neighbor (NN) hopping within the superconductor and V_{SC} is the strength of the local on-site attractive interaction between opposite-spin electrons that drives Cooper pairing.

Normal-superconductor interface: The ring contains two NS interfaces connecting the last site of the normal region to the first site of the SC region and the last SC site back to the first normal site. The coupling between the normal and the superconducting regions is described by

$$H_{NS} = -t_{NS} \sum_{\langle i,j \rangle_{\text{int}}} (c_i^\dagger c_j + h.c.), \quad (5)$$

where t_{NS} is the hopping amplitude at the NS interfaces. For perfectly transparent interfaces, $t_{NS} = t_N = t_{SC}$.

B. The Method

1. Mean-Field BdG Hamiltonian

The quartic interaction term in H_{SC} is decoupled using the Bogoliubov mean-field approximation^{55–58}. We introduce the spatially resolved superconducting order parameter (OP)

$$\Delta_i \equiv \langle c_{i\downarrow} c_{i\uparrow} \rangle, \quad (6)$$

which represents the local pair amplitude at site i . Since the pairing interaction is absent in the normal region ($V_i = 0$ for $i \in N$), $V_i \Delta_i$ is strictly zero there, even though Δ_i itself can be nonzero due to the superconducting proximity effect. The interaction also generates

a Hartree shift in the on-site energy of the superconducting region which is

$$U_i^H = -V_i \sum_{E_n < 0} |u_{i,n}|^2, \quad (7)$$

where $u_{i,n}$ is the particle (electron-like) amplitude at site i of quasiparticle eigenstate n with energy E_n , and the sum runs over all negative-energy states. Here $V_i = V_{SC}$ for $i \in SC$ and $V_i = 0$ for $i \in N$, so the Hartree shift is nonzero only in the SC region. Combining the AAH potential and the Hartree shift into the total effective onsite energy, we get

$$U_i^{\text{tot}} = U_i^{\text{AAH}} + U_i^H, \quad (8)$$

where U_i^{AAH} is the on-site AAH potential. The mean-field Hamiltonian reads as

$$H_{\text{MF}} = \sum_{i,\sigma} U_i^{\text{tot}} c_{i\sigma}^\dagger c_{i\sigma} - \sum_{\langle i,j \rangle, \sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + \sum_{i \in SC} (V_i \Delta_i c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger + h.c.), \quad (9)$$

where t_{ij} takes the value t_A , t_B , t_{SC} , or t_{NS} according to whether the bond (i,j) lies entirely within the normal region, entirely within the SC region, or crosses an NS interface, respectively.

2. Bogoliubov-de Gennes method

The Hamiltonian H_{MF} is diagonalized via the Bogoliubov transformation

$$\begin{aligned} c_{i\uparrow} &= \sum_n (u_{i,n} \gamma_{n\uparrow} - v_{i,n}^* \gamma_{n\downarrow}^\dagger), \\ c_{i\downarrow} &= \sum_n (u_{i,n} \gamma_{n\downarrow} + v_{i,n}^* \gamma_{n\uparrow}^\dagger), \end{aligned} \quad (10)$$

where $\gamma_{n\sigma}^\dagger$ ($\gamma_{n\sigma}$) creates (annihilates) a Bogoliubov quasiparticle in state n with spin σ ($\uparrow, \downarrow$), and $u_{i,n}$, $v_{i,n}$ are the particle and hole amplitudes at site i in eigenstate n . Expressing H_{MF} in the Nambu spinor basis $\Psi = (c_{1\uparrow}, \dots, c_{L\uparrow}, c_{1\downarrow}^\dagger, \dots, c_{L\downarrow}^\dagger)^T$ yields the $(2L \times 2L)$ BdG matrix

$$\mathcal{H}_{\text{BdG}} = \begin{pmatrix} \hat{K} & \hat{\Delta} \\ \hat{\Delta}^\dagger & -\hat{K} \end{pmatrix}, \quad (11)$$

where the $(L \times L)$ blocks are defined element-wise as

$$K_{ij} = U_i^{\text{tot}} \delta_{ij} - t_{ij}(1 - \delta_{ij}), \quad (12)$$

$$\Delta_{ij} = V_i \Delta_i \delta_{ij}. \quad (13)$$

Here δ_{ij} is the Kronecker delta. The diagonal of $\hat{K}$ contains the effective on-site energies U_i^{tot} , while its off-diagonal elements encode the NN hopping t_{ij} between neighboring sites. The pairing block $\hat{\Delta}$ is diagonal and nonzero only within the SC region, where it equals $V_{SC} \Delta_i$, and it is identically zero in the normal region

because $V_i = 0$ there. The particle-hole symmetry of the BdG formalism is reflected in the hole block $-\hat{K}$, whose sign is opposite to the particle block. The BdG equation

$$\mathcal{H}_{\text{BdG}} \begin{pmatrix} \mathbf{u}_n \\ \mathbf{v}_n \end{pmatrix} = E_n \begin{pmatrix} \mathbf{u}_n \\ \mathbf{v}_n \end{pmatrix} \quad (14)$$

yields $2L$ eigenpairs $\{E_n, (\mathbf{u}_n, \mathbf{v}_n)\}$, where $\mathbf{u}_n = (u_{1,n}, \dots, u_{L,n})^T$ and $\mathbf{v}_n = (v_{1,n}, \dots, v_{L,n})^T$. Particle-hole symmetry guarantees that eigenvalues appear in $\pm E_n$ pairs, it suffices to sum over states with $E_n < 0$ in the self-consistency equations below.

3. Self-Consistency equations

The OP Δ_i and the Hartree shift U_i^H are obtained through an iterative self-consistent procedure^{8,9}. Starting from an initial guess $\Delta_i^{(0)} = 1$ for $i \in SC$, $\Delta_i^{(0)} = 0$ for $i \in N$, and $U_i^{H,(0)} = 0$ everywhere, at each iteration k we diagonalize $\mathcal{H}_{\text{BdG}}$ and update the OP and Hartree shift as

$$\Delta_i^{(k+1)} = - \sum_{E_n < 0} u_{i,n} v_{i,n}^*, \quad (15)$$

$$U_i^{H,(k+1)} = -V_i \sum_{E_n < 0} |u_{i,n}|^2, \quad (16)$$

where the sums are taken over all quasiparticle eigenstates with $E_n < 0$, computed using the values $\Delta_i^{(k)}$ and $U_i^{H,(k)}$ from the previous step. This procedure is repeated until the solution converges, as described below.

4. Convergence

We monitor convergence by checking the relative change in Δ_i between successive iterations^{8,9}. At each site i , we set

$$\frac{|\Delta_i^{(k+1)} - \Delta_i^{(k)}|}{|\Delta_i^{(k+1)}|} < 10^{-3}. \quad (17)$$

Once Eq. (17) holds at every relevant site, the loop is stopped and the converged $|\Delta_i|$ is taken as the physical result.

III. NUMERICAL RESULTS AND DISCUSSION

A. AAH Normal Region

We express all energies in units of electron-volt (eV) throughout the work. The hopping amplitudes are set to $t_N = t_B = t_{SC} = t_{NS} = 1$ and the attractive Hubbard interaction strength is taken as $V_{SC} = 1.5$.

To gain insight into the proximity effect within the AAH region, we first explore the spatial distribution of

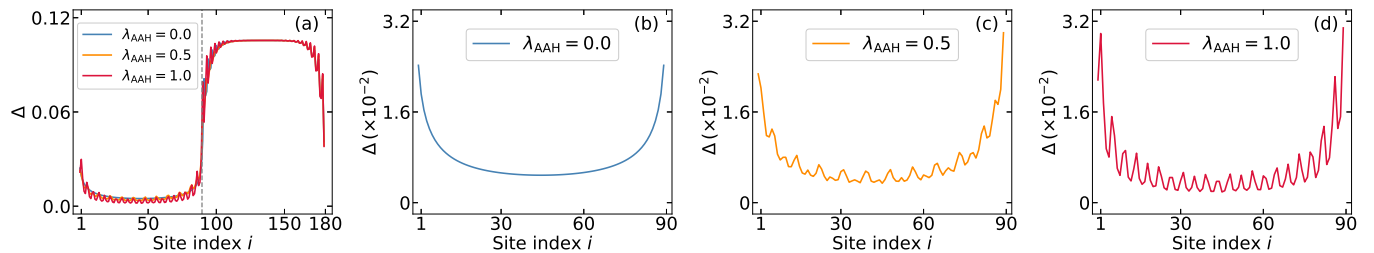


FIG. 2: (Color online.) Spatial profile of the self-consistently determined pairing amplitude Δ_i as a function of the site index i for three distinct values of the quasiperiodic potential strength λ_{AAH} . Panel (a) shows the order parameter over the entire hybrid ring for $\lambda_{\text{AAH}} = 0, 0.5$, and 1 . Panels (b), (c), and (d) present enlarged views of the normal region for $\lambda_{\text{AAH}} = 0, 0.5$, and 1 , respectively, highlighting the evolution of the spatial fluctuations of the induced pairing amplitude with increasing quasiperiodic potential strength. The dashed vertical line indicates the normal-superconductor interface. The system parameters are $L = 180$, $L_N = 90$, $L_{SC} = 90$, $t_N = t_{SC} = t_{NS} = 1$, $V_{SC} = 1.5$, and $\phi = 0$.

the self-consistently determined pairing amplitude. Figure 2 shows the real-space profile of OP Δ_i as a function of site index i for three different values of the AAH potential strength $\lambda_{\text{AAH}} = 0, 0.5, 1$, all well within the extended phase ($\lambda_{\text{AAH}} < 2t_N = 2$). The spatial variation of the order parameter in the normal region is shown separately in Figs. 2(b)-(d) for the mentioned three values of the quasiperiodic modulation strength, so that the spatial fluctuations within each region can be inspected simultaneously without the curves overlapping. We discuss the behavior in each region separately.

In the SC region, the order parameter attains a large, approximately uniform bulk value in the interior of the SC chain, reflecting the self-consistent pairing driven by the attractive on-site interaction V_{SC} . The bulk value is determined primarily by the pairing interaction V_{SC} and the density of states of the SC region, and remains essentially unchanged across the three values of λ_{AAH} studied here, since the AAH potential acts only in the normal region and does not directly affect the SC pairing in the bulk. Close to the NS interfaces, however, the induced pairing amplitude is visibly reduced from its bulk value. This suppression originates from the inverse proximity effect, where electrons from the normal region penetrate into the superconductor and locally weaken the pairing correlations. Superimposed on the suppression near the interfaces, small oscillations in Δ_i are visible, reminiscent of Friedel oscillations arising from the interference of quasiparticle states reflected at the NS interface. These features are present for all three values of λ_{AAH} and become slightly more pronounced as λ_{AAH} increases, consistent with a stronger electronic mismatch between the normal and SC regions at larger quasiperiodic potential strengths.

In the normal region, the order parameter Δ_i is proximity-induced. Since the pairing interaction is absent ($V_i = 0$) in this region, $V_i \Delta_i$ becomes zero, and the nonzero values of Δ_i reflect the pairing amplitude $\langle c_{i\downarrow} c_{i\uparrow} \rangle$ leaking from the SC through the NS interfaces via the Andreev processes. The proximity-induced pairing amplitude within the AAH region is significantly smaller in magnitude than the bulk value in SC region, as expected. The spatial profile of Δ_i in the normal region is not smooth but exhibits site dependent fluctuations,

whose amplitude increases with λ_{AAH} . These fluctuations arise from the quasiperiodic modulation of the on-site potential, which modifies the spatial structure of the single-particle wave functions and, consequently the local pairing amplitude. For $\lambda_{\text{AAH}} = 0$ (blue curve) the profile is nearly smooth, consistent with extended plane-wave-like eigenstates. As λ_{AAH} increases (orange and red curves), the stronger quasiperiodic modulation of the wave functions leads to more pronounced spatial variations in pairing amplitude. At the same time, the overall magnitude of the induced order parameter gradually decreases as λ_{AAH} is increased. The quasiperiodic potential progressively reduces the ability of Cooper pairs to propagate through the normal region by suppressing the spatial extent of the single particle states participating in the proximity effect. Consequently, superconducting correlations penetrate less efficiently into the AAH segment, leading to a reduction of the induced pair amplitude throughout the normal region. Therefore, increasing quasiperiodicity has a dual effect. It enhances the spatial inhomogeneity of the induced superconducting correlations while simultaneously reducing their strength. These real-space characteristics provide the microscopic origin of the behavior observed in the subsequent analysis of the induced order parameter as a function of the quasiperiodic modulation strength.

To quantify this suppression and relate it to the localization transition of the AAH chain we analyze the order parameter Δ_i at two different lattice sites ($i = 21, 46$) in the normal region as a function of λ_{AAH} . One can see from Fig. 3(a) that Δ_i decreases monotonically as the quasiperiodic modulation strength is increased, and drops to essentially to zero once λ_{AAH} crosses the critical value $2t_N$. This is a direct indication that the proximity effect is completely diminished once the normal-region states become localized since the leakage of the Cooper-pair into the normal region reduces as λ_{AAH} drives the system from extended to the localized region. To understand the mechanism behind this suppression, we plot the energy eigenvalues as a function of λ_{AAH} in Fig. 3(b), which reflects the progressive fragmentation of the AAH spectrum from a continuous band in the extended phase to a fractal-like structure at the critical point, as $\lambda_{\text{AAH}} \rightarrow 2t_N$. We further investigate the con-

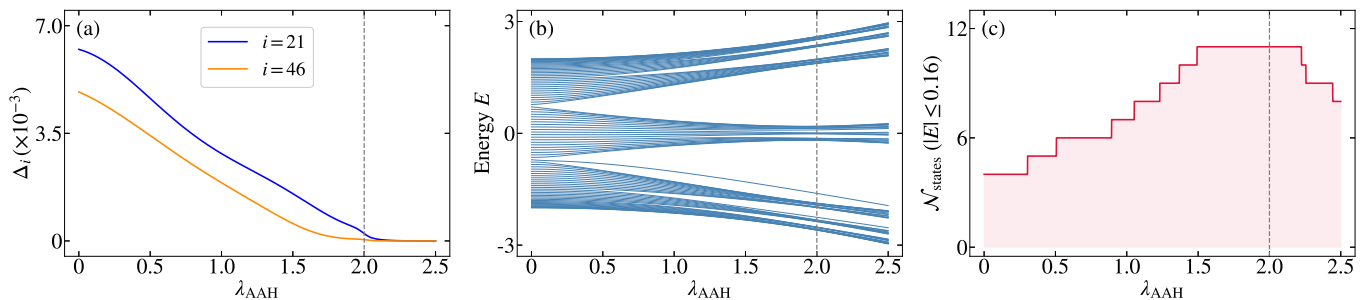


FIG. 3: (Color online.) (a) Self-consistently determined superconducting order parameter Δ_i at two representative sites ($i = 21$ and $i = 46$) in the normal region as a function of the quasiperiodic potential strength λ_{AAH} . The vertical dashed line at $\lambda_{\text{AAH}} = 2t_N$ marks the localization transition of the AAH chain. (b) Energy spectrum of the finite AAH chain with $L_N = 90$ sites as a function of λ_{AAH} , illustrating the evolution of the eigenstates across the localization transition. (c) Number of quasiparticle states, $\mathcal{N}_{\text{states}}$, with energies lying within the window $[-0.16, 0.16]$ as a function of λ_{AAH} . The remaining system parameters are the same as in Fig. 2.

sequences of the spectral restructuring on the induced order parameter, and for that we compute the number of states $\mathcal{N}_{\text{states}}$ with energies in the window $[-0.16, 0.16]$ as a function of λ_{AAH} . The results are shown in Fig. 3(c). This energy window corresponds directly to the energy scale $V_{\text{SC}}\Delta_{\text{mid}}$, where Δ_{mid} is the self-consistently determined pairing amplitude evaluated at the center of the SC region. This allows us to evaluate how many states are available to participate in the pairing. Remarkably, $\mathcal{N}_{\text{states}}$ increases steadily with λ_{AAH} and saturates in the vicinity of the critical point capturing dense accumulation of the multifractal states. For $\lambda_{\text{AAH}} > 2t_N$, however, $\mathcal{N}_{\text{states}}$ decreases again as the system enters the localized phase. In this regime, the eigenstates become exponentially localized, reducing their spatial overlap and thereby limiting their ability to support proximity-induced pairing correlations.

The simultaneous vanishing of Δ_i and growth of $\mathcal{N}_{\text{states}}$ establishes that the suppression of the proximity effect is *not* due to a depletion of states around the Fermi energy. Rather, it is a consequence of the *spatial* localization of these states. If the density of states were the controlling factor, one would expect a larger number of states to enhance the induced pairing correlations. The opposite trend is observed here. Although more states accumulate within the superconducting energy window as λ_{AAH} increases, their ability to participate in the proximity effect is progressively reduced. The reason lies in the localization properties of the eigenstates. The proximity-induced pairing amplitude is governed by the self-consistency equation described in Eq. (15) in which the contribution of each quasiparticle state n is weighted by the product of its particle and hole amplitudes $u_{i,n}$ and $v_{i,n}^*$ at site i . In the localized phase, eigenstates are exponentially confined with localization length $\xi \sim 1/\ln(\lambda_{\text{AAH}}/2t_N)$ ¹⁸, so that states localized away from site i contribute negligibly to Δ_i regardless of their energy. The growing number of sub-gap states therefore fails to enhance the proximity effect because these states have exponentially small amplitude at i . The suppression of Δ_i thus reflects the dominance of spatial localization over the spectral accumulation of low-energy states, and constitutes

a clear real-space, self-consistent BdG signature of the AAH localization-delocalization transition.

To characterize the sensitivity of the proximity effect to the specific configurations of the quasiperiodic potential, we track the self-consistently determined order param-

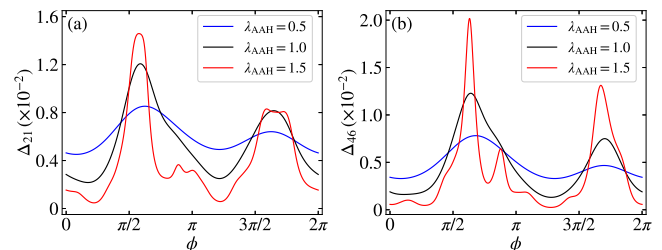


FIG. 4: (Color online.) Variation of the order parameter Δ_i at sites (a) $i = 21$ and (b) $i = 46$ as a function of the AAH phase $\phi \in [0, 2\pi]$ for three distinct values of the quasiperiodic potential strength $\lambda_{\text{AAH}} = 0.5, 1.0$, and 1.5 . The remaining system parameters are the same as in Fig. 2.

eter at two sites $i = 21$ and 46 as the AAH phase ϕ is varied which is shown in Figs. 4(a) and (b) respectively. For a chain of fixed length L_N , sweeping $\phi \in [0, 2\pi]$ generates a continuous family of quasiperiodic potentials that are statistically equivalent in the thermodynamic limit but differ locally in their on-site energy landscapes for any finite L_N . Thus, varying ϕ continuously samples different configurations of the quasiperiodic potential, and the resulting curve $\Delta_i(\phi)$ directly captures the sample-to-sample fluctuations of the proximity-induced pairing amplitude at fixed disorder strength λ_{AAH} . For all the three cases, $\Delta_i(\phi)$ fluctuates smoothly with ϕ , which is a direct signature of the finite-size effect. Notably, the amplitude of these fluctuations grows with increasing λ_{AAH} , indicating that the proximity-induced pairing amplitude becomes increasingly sensitive to the specific configuration of the quasiperiodic potential as the quasiperiodic modulation strength increased. While the fluctuations across different values of ϕ remain relatively modest in the weak to moderate quasiperiodic regime, they become substantially more pronounced at stronger disorder, reflecting a growing dependence of the local pairing ampli-

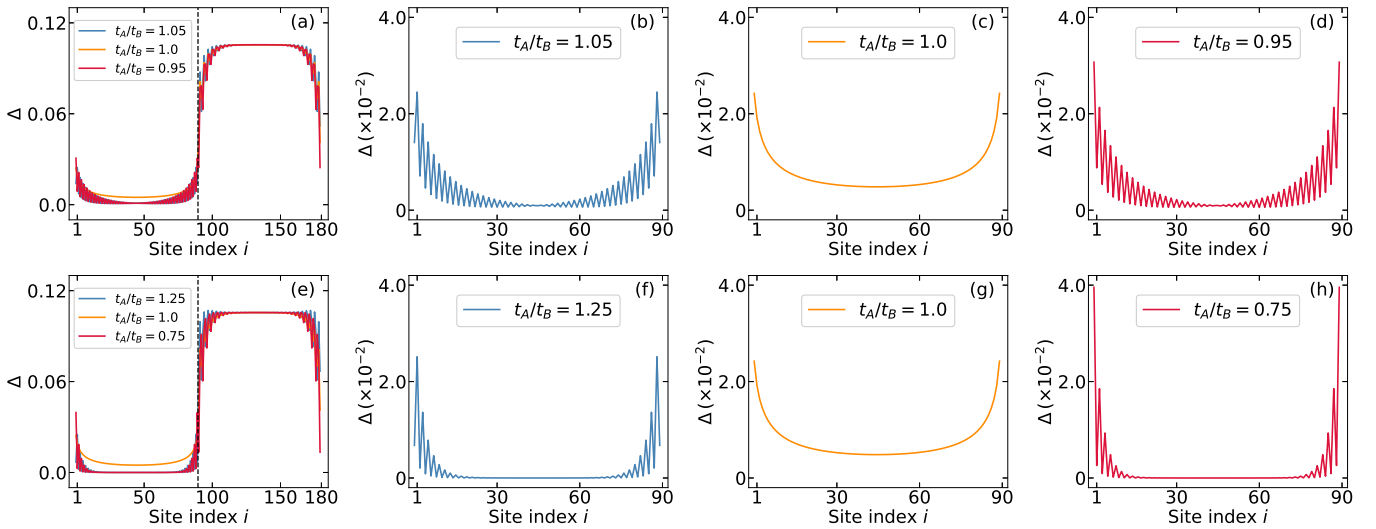


FIG. 5: (Color online.) Real-space profile of the self-consistently determined superconducting pairing amplitude Δ_i as a function of the site index i for different values of the SSH dimerization ratio t_A/t_B . Panels (a) and (e) show the variation of the order parameter over the entire hybrid ring for the weakly and strongly dimerized cases, respectively. Panels (b) and (d) display enlarged views of the normal region for the weakly dimerized cases with $t_A/t_B = 1.05$ and 0.95 . Similarly, panels (f) and (h) show the strongly dimerized cases with $t_A/t_B = 1.25$ and 0.75 . Whereas panel (c) and (g) represent the uniform chain ($t_A/t_B = 1$) for comparison. The dashed vertical line indicates the normal-superconductor interface. The system parameters are $L = 180$, $L_N = 90$, $L_{SC} = 90$, $t_{SC} = t_{NS} = 1.0$, and $V_{SC} = 1.5$.

tude on the fine details of the on-site energy landscape.

B. SSH Normal Region

Having studied the AAH case, we now turn to the SSH model and investigate its impact on the proximity-induced superconducting correlations. The spatial distribution of the self-consistent order parameter Δ_i for different hopping ratios t_A/t_B is shown in Fig. 5. Panels (a) and (e) of Fig. 5 display the spatial variation of the order parameter over the entire hybrid ring for the weakly and strongly dimerized cases, respectively. The corresponding enlarged views shown in Figs. 5 (b)-(d) and (f)-(h) focus on the normal segment, making the evolution of the proximity-induced pairing correlations easier to visualize.

For the uniform chain, $t_A/t_B = 1$, the order parameter decreases smoothly from the NS interfaces towards the middle of the normal region. Although the induced pairing is weakest near the center, it remains finite throughout the chain, indicating that superconducting correlations can penetrate relatively far into the normal segment.

As soon as a small dimerization is introduced, the smooth profile gives way to an oscillatory one. From Figs. 5(b)-(d) we can see that in the weak dimerization case for both $t_A/t_B = 1.05$ and 0.95 , where clear oscillations appear near the interfaces and continue well into the interior of the chain. The overall envelope of the induced pairing is not very different from the uniform case, but its spatial modulation is noticeably modified. The oscillations survive over a large part of the normal region before gradually diminishing towards the center, suggesting that weak dimerization alters the distribution of the

superconducting correlations without strongly suppressing them. A stronger effect is observed for $t_A/t_B = 1.25$ and 0.75 which is shown in Figs. 5(f)-(h). In these cases, the oscillations are concentrated near the interfaces and decay much more rapidly with distance. As a result, the induced pairing becomes extremely small in the middle of the chain. Compared with the weakly dimerized cases, the range over which superconducting correlations extend into the normal region is significantly reduced. The proximity effect therefore becomes increasingly confined to the vicinity of the NS boundaries as the hopping imbalance is increased. In this regime, the SSH chain behaves as a poor medium for Cooper pair leakage from the SC region. The strong alternation between t_A and t_B effectively suppresses transport through the bulk, leading to a reduced penetration depth of superconducting correlations.

To get a more quantitative handle on how dimerization influences the proximity effect, we fix two sites in the normal region, $i = 20$ and $i = 40$, and track the induced order parameter Δ_i at those sites as we sweep the dimerization ratio t_A/t_B . The result is shown in Fig. 6(a). What stands out immediately is how sharply Δ_i peaks at $t_A/t_B = 1$ and how quickly it falls away on either side. The site closer to the NS interface ($i = 20$) carries a larger induced pairing than the deeper site ($i = 40$), as one would expect, but both sites tell the same story. This behavior is a direct consequence of the bulk energy gap of the SSH chain defined as

$$E_{\text{gap}} = 2t_B \left| 1 - \frac{t_A}{t_B} \right|, \quad (18)$$

which closes at $t_A/t_B = 1$ and grows linearly on either side which can be seen from Fig. 6(b). The proximity-

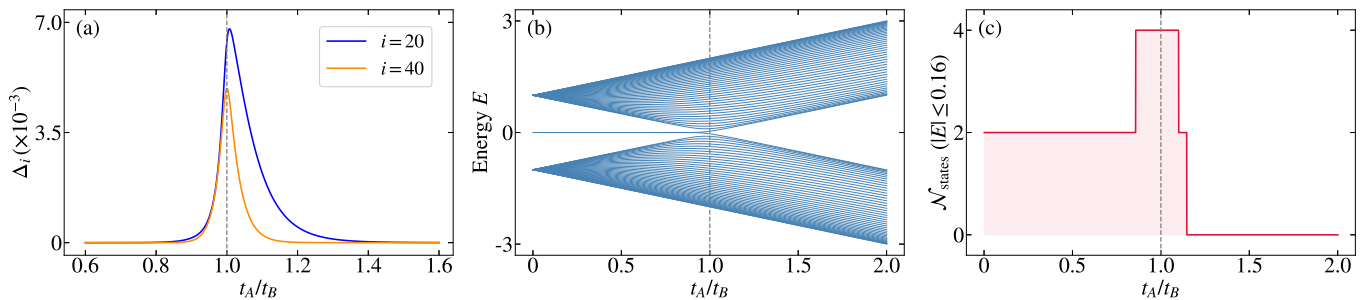


FIG. 6: (Color online.) (a) Self-consistently determined superconducting order parameter Δ_i at two representative sites ($i = 20$ and $i = 40$) in the normal region as a function of the SSH dimerization ratio t_A/t_B . The induced pairing reaches its maximum near $t_A/t_B = 1.0$, where the SSH chain undergoes a topological phase transition. (b) Energy spectrum of the finite SSH chain with $L_N = 90$ sites as a function of the dimerization ratio t_A/t_B , showing the evolution of the bulk bands and the gap closing at $t_A/t_B = 1$. (c) Number of states, $\mathcal{N}_{\text{states}}$, with energies lying within the window $[-0.16, 0.16]$ as a function of t_A/t_B . The remaining system parameters are the same as in Fig. 5.

induced order parameter is dominated by quasiparticle states near the Fermi level, which carry the largest mixed particle-hole character as measured by the product $u_{i,n}v_{i,n}^*$ in the self-consistency equation. When $t_A/t_B = 1$, the SSH energy spectrum is gapless which means there are large number of energy labels available near the energy which can also be seen from Fig. 6(c) that within the low-energy window $|E| \leq 0.16$ the count hits its maximum. These are the states which actually do the work for carrying the Cooper pairs from SC region to the SSH region. So having more of them naturally makes the proximity effect stronger. The moment t_A/t_B moves away from unity, the spectrum becomes gapped and the number of states within that energy windows drops off with the gap opening shown in the Fig. 6(b). It does not open gradually, it grows quite quickly as the dimerization strengthens, and it leaves fewer and fewer states available to mediate the pairing which can also be seen from Fig. 6(c). So, when the SSH gap is large, no such states exist within the relevant energy window and the induced superconducting correlations is suppressed. The observed variation of Δ_i is therefore serves as a real-space, self-consistent BdG signature of the SSH topological phase transition.

C. SSH-AAH Normal Region

We now inspect the combined effect of the SSH dimerization and the AAH quasiperiodic potential on the proximity-induced pairing correlations. The spatial variations of the self-consistent order parameter for several SSH configurations and different values of λ_{AAH} is shown in Fig. 7. Panels (a)-(e) show the order parameter over the entire hybrid ring, while Figs. 7(a1)-(a3), (b1)-(b3), (c1)-(c3), (d1)-(d3), and (e1)-(e3) provide enlarged views of the normal segment. Within each set of enlarged panels, the results correspond to $\lambda_{\text{AAH}} = 0, 0.5$, and 1 , respectively.

The first row shown in Figs. 7(a1)-(a3) corresponds to the uniform chain ($t_A = t_B = 1$). In the absence of the AAH potential, the induced order parameter varies

smoothly across the normal region and remains finite throughout its length. As λ_{AAH} is increased, this smooth profile gradually develops pronounced site-to-site variations. Although superconducting correlations still penetrate the entire normal segment, their magnitude becomes increasingly nonuniform, and the induced pairing amplitude in the bulk decreases.

The weakly dimerized chains represent an intermediate regime between the uniform and strongly dimerized limits are shown in Figs. 7(b1)-(b3) for $t_A/t_B = 1.05$ and Figs. 7(c1)-(c3) for $t_A/t_B = 0.95$. In both cases the induced order parameter exhibits clear oscillations that originate near the normal-superconductor interfaces and persist over a large part of the normal region. Unlike the strongly dimerized case, these oscillations are not confined to the interface but extend deep into the bulk, indicating that superconducting correlations can still propagate over long distances. As the quasiperiodic potential strength λ_{AAH} is increased, the oscillatory pattern becomes increasingly irregular and its amplitude is gradually suppressed. Consequently, the combined effect of weak dimerization and quasiperiodic modulation leads to a reduction in both the coherence and the penetration depth of the proximity-induced superconducting correlations.

A distinctly different behavior is observed for the strongly dimerized chains which is shown in Figs. 7(d1)-(d3) and (e1)-(e3). For $t_A = 1.25, t_B = 1$ and $t_A = 1, t_B = 1.25$, the induced order parameter exhibits pronounced oscillations only in the vicinity of the NS interfaces. Unlike the weakly dimerized case, these oscillations decay rapidly and do not extend into the bulk of the normal region, leaving only a very small pairing amplitude near the center of the chain. Upon introducing the quasiperiodic AAH potential, the oscillatory pattern close to the interfaces becomes more irregular, while the residual superconducting correlations in the bulk are further suppressed. Since the penetration of Cooper pairs is already strongly limited by the large hopping imbalance, the overall reduction of the proximity effect in this regime is governed primarily by the SSH dimerization, with the quasiperiodic potential providing an additional but com-

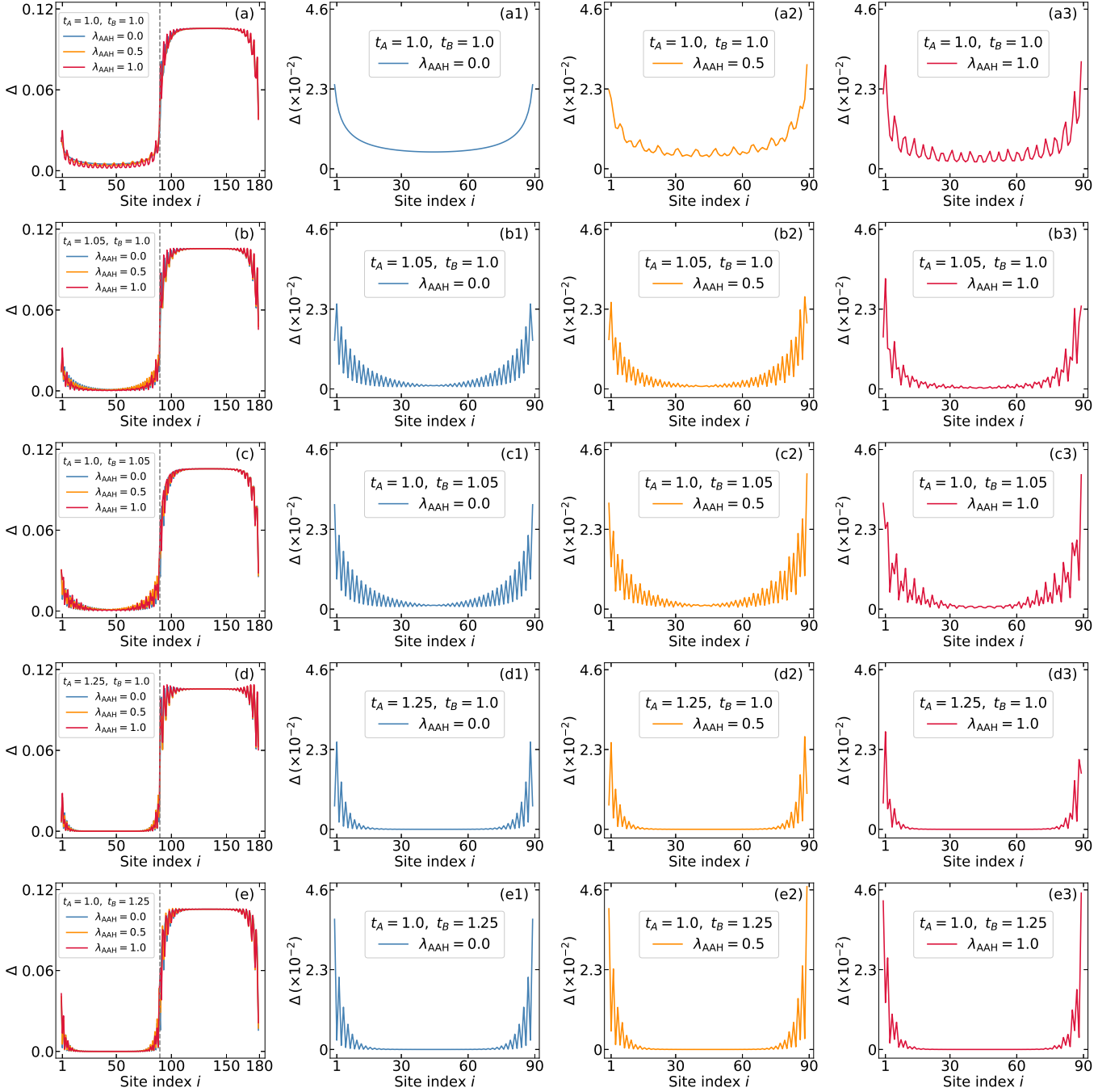


FIG. 7: (Color online.) Self-consistently determined real-space profile of the pairing amplitude Δ_i as a function of the site index i for different combinations of the SSH dimerization ratio t_A/t_B and the AAH quasiperiodic potential strength λ_{AAH} . In all cases, results are shown for $\lambda_{AAH} = 0, 0.5$ and 1 . Panels (a)-(e) display the spatial variation of Δ_i over the full hybrid ring for the uniform chain ($t_A/t_B = 1$), the weakly dimerized cases ($t_A/t_B = 1.05$ and 0.95), and the strongly dimerized cases ($t_A/t_B = 1.25$ and 0.75), respectively. Panels (a1)-(a3) show enlarged views of the normal region for the uniform chain. Panels (b1)-(b3) and (c1)-(c3) correspond to the weakly dimerized cases with $t_A > t_B$ and $t_A < t_B$, respectively, while panels (d1)-(d3) and (e1)-(e3) correspond to the strongly dimerized cases with $t_A > t_B$ and $t_A < t_B$, respectively. The dashed vertical lines mark the NS interfaces. The system parameters are $L = 180$, $L_N = 90$, $L_{SC} = 90$, $t_{SC} = t_{NS} = 1$, $V_{SC} = 1.5$, and $\phi = 0$.

paratively weaker suppression. The degree of dimerization primarily determines how far the proximity-induced pairing correlations can penetrate into the normal region and how they oscillate spatially, while the quasiperiodic potential mainly introduces spatial fluctuations and suppresses the magnitude of the induced pairing. When both

are present, the resulting order-parameter profile reflects the competition between these two tendencies, leading to increasingly superconducting correlations as either the dimerization strength or λ_{AAH} is increased.

IV. CLOSING REMARKS AND OUTLOOK

The present work explores the superconducting proximity effect in a 1D hybrid ring by considering different realizations of the normal region within a self-consistent Bogoliubov-de Gennes framework. By studying the diagonal AAH model, SSH model, and the combined SSH-AAH model, we investigate how quasiperiodicity and hopping dimerization influence the propagation of superconducting correlations into the normal segment.

For the diagonal AAH model, increasing the quasiperiodic potential strength gradually suppresses the induced order parameter in the normal region while simultaneously enhancing its spatial fluctuations. The AAH phase dependence of the order parameter shows that different quasiperiodic realizations can exhibit noticeable variations in the induced pairing for finite systems.

For the SSH model, the dimerization ratio plays a vital role in determining the spatial profile of the induced pairing correlations. In the uniform limit, the proximity-induced pairing extends throughout the normal region with a smooth spatial variation. The weak dimerization introduces oscillatory modulations in the order parameter, and these oscillations persist over a large portion of the normal region, indicating that superconducting correlations can still penetrate deep into the bulk. As the dimerization strength increases, the oscillations become confined to regions near the normal-superconductor interfaces and decay rapidly away from the boundaries. Consequently, the penetration depth of the proximity effect is significantly reduced, leaving only a negligible induced pairing in the central part of the chain. The topological phase transition of the SSH chain also reflected in the behavior of proximity induced pairing.

The combined SSH-AAH model reveals the interplay between hopping dimerization and quasiperiodic modulation. The SSH hopping pattern determines the overall oscillatory character and penetration of the induced pairing, while the AAH potential introduces additional spatial fluctuations and further suppresses its magnitude. In the weakly dimerized regime, the oscillatory pairing profile survives deep inside the normal region but becomes increasingly irregular as the quasiperiodic potential strength is increased. In contrast, for strong dimerization, where the induced pairing is already localized near the interfaces, the AAH potential mainly enhances the suppression of the residual superconducting correlations.

Overall, the present results demonstrate that hopping dimerization and quasiperiodicity provide two complementary mechanisms for controlling proximity-induced

superconductivity in one-dimensional hybrid systems. While dimerization governs the penetration depth and oscillatory nature of the induced pairing, the quasiperiodic potential controls its spatial inhomogeneity and overall magnitude. Their combined effect offers a flexible route for engineering superconducting correlations in low-dimensional quasiperiodic structures and may be useful for designing hybrid devices with tunable superconducting properties. Future investigations may explore the effects of finite temperature, and longer-range hopping. It would also be interesting to study circular charge and spin currents by applying magnetic flux through the hybrid ring.

The experimental realization of quasiperiodic systems has advanced considerably in recent years. Progress in quasicrystal growth⁵⁹, the fabrication of quasiperiodic electronic structures such as Pb monolayers on quasicrystalline substrates⁶⁰, the atom-by-atom construction of synthetic electronic quasicrystals based on Penrose tilings⁶¹, the observation of superconductivity in the icosahedral Al-Zn-Mg quasicrystal⁴⁸, and the recent realization of a superconducting moiré quasicrystal⁶² suggest that superconducting hybrid devices incorporating quasiperiodic materials are within experimental reach. These developments provide a promising platform for directly testing the proximity-induced superconducting correlations predicted in the present work and, more generally, for exploring superconducting proximity effects in quasiperiodic systems.

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DATA AVAILABILITY STATEMENT

The data supporting the findings of this study are included in the manuscript.

DECLARATION

Conflict of interest The authors declare no conflict of interest.

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